Skills Summary

Statistical Questions and Describing Variability

Use this reference while completing your worksheet or when studying.

Telling a Statistical Question Apart

Rule: a question is **statistical** when answering it means collecting data whose values *vary*. One named thing at one named moment has one answer, so it is not statistical.

Test: could two honest answers differ? If yes, it is statistical, and the range of the collected values is one way to show how much they vary.

Example: "How old is the oldest tree in the park?" or "How old are the trees in the park?" — which is statistical? Five trees were measured: 8, 11, 9, 14, 12 years.
Step 1. The first question names one tree and has a single answer; the second is answered by collecting an age from every tree, and those ages differ, so the second is the statistical one.
Step 2. The variability is real and can be measured: the ages run from 8 to 14, a range of $14 - 8 = \mathbf{6}$ years.

Try it: Which is statistical — "how many pages did Sam read today?" or "how many pages do the students read each day?" — and what is the range of 15, 22, 18, 27, 20 pages?

check: second question; range **12**

Measuring Spread: Range and Interquartile Range

Rule: range = largest − smallest. It uses only the two end values, so one unusual value changes it completely.

Interquartile range = upper quartile − lower quartile. Put the values in order, split them into a lower half and an upper half, and take the median of each half. It measures the width of the middle half of the data, so end values do not move it.

Example: Twelve values in order: 1, 3, 4, 6, 8, 9, 11, 13, 14, 16, 18, 20. Find the quartiles and the interquartile range.
Step 1. Twelve values split evenly into a lower six and an upper six, so take the median of each half.
Step 2. Lower half 1, 3, 4, 6, 8, 9 gives **5**; upper half 11, 13, 14, 16, 18, 20 gives **15**; the interquartile range is $15 - 5 = \mathbf{10}$.

Try it: Find the interquartile range of 2, 4, 6, 8, 12, 14, 18, 20.

check: **11**

Describing a Data Set: Centre and Spread Together

Rule: a description needs **a typical value** and **how spread out** the values are. Two sets can share a mean and look nothing alike.

Choosing the centre: the mean is pulled towards an unusually large or small value; the median is not. When one value sits far from the rest, the median describes a typical member better.

Example: Nine deliveries took 2, 3, 3, 4, 4, 5, 6, 9, 36 minutes. Which measure describes a typical delivery?

Step 1. One value, 36 minutes, sits far above the rest, so compute both the mean and the median and see how far apart they land.

Step 2. The times total 72, so the mean is $72 \div 9 = \mathbf{8}$ minutes, while the median (the fifth value) is **4** minutes — the median, because seven of the nine deliveries took under 8 minutes.

Try it: Find the median of 1, 2, 2, 3, 3, 4, 5, 8, 26, and say whether it or the mean better describes a typical value.

check: median **3**, and it describes it better

(!) Watch out: an equal mean does not mean equal data. Two teams averaging 22 points can have ranges of 4 and 24 — the spread is the part that tells them apart.